PlanBench-XL
Mean ± SD

	Qwen2.5-7B		Llama-3.1-8B		Qwen3.8-27B	
Method	**Stage 0**	**Stage 1**	**Stage 0**	**Stage 1**	**Stage 0**	**Stage 1**
Frozen Base	0.2073 ±0.0357	0.2073 ±0.0357	0.2232 ±0.0344	0.2232 ±0.0344	0.2554 ±0.0411	0.2554 ±0.0411
AREW / T3	0.2073 ±0.0357	0.2232 ±0.0344	0.2232 ±0.0344	0.2232 ±0.0344	0.2554 ±0.0411	0.2554 ±0.0411
GiGPO	0.2073 ±0.0357	0.2232 ±0.0344	0.2232 ±0.0344	0.2232 ±0.0344	0.2554 ±0.0411	0.2709 ±0.0550
InfoPO	0.2073 ±0.0357	0.2436 ±0.0359	0.2232 ±0.0344	0.2554 ±0.0411	0.2554 ±0.0411	0.3350 ±0.0281
WorldEvolver	0.2073 ±0.0357	0.4036 ±0.0217	0.2232 ±0.0344	0.4323 ±0.0206	0.2554 ±0.0411	0.5359 ±0.0199
PoS / EnvProbe	0.2073 ±0.0357	**0.5250** ±0.0111	0.2232 ±0.0344	**0.5690** ±0.0087	0.2554 ±0.0411	**0.6757** ±0.0063
Step-DPO	0.2073 ±0.0357	0.3104 ±0.0605	0.2232 ±0.0344	0.3554 ±0.0743	0.2554 ±0.0411	0.4713 ±0.0744
Outcome-PG	0.2073 ±0.0357	0.2273 ±0.0410	0.2232 ±0.0344	0.2232 ±0.0344	0.2554 ±0.0411	0.2750 ±0.0618
Outcome-Value	0.2073 ±0.0357	0.2232 ±0.0344	0.2232 ±0.0344	0.2313 ±0.0281	0.2554 ±0.0411	0.2832 ±0.0529
Outcome-RLOO	0.2073 ±0.0357	0.2273 ±0.0410	0.2232 ±0.0344	0.2232 ±0.0344	0.2554 ±0.0411	0.2832 ±0.0529
Info-RL Neighbor	0.2073 ±0.0357	0.2232 ±0.0344	0.2232 ±0.0344	0.2436 ±0.0359	0.2554 ±0.0411	0.2954 ±0.0432
Unstratified	0.2073 ±0.0357	0.2313 ±0.0404	0.2232 ±0.0344	0.2477 ±0.0531	0.2554 ±0.0411	0.3381 ±0.0666
Unweighted	0.2073 ±0.0357	0.2780 ±0.0504	0.2232 ±0.0344	0.2751 ±0.0726	0.2554 ±0.0411	0.4147 ±0.0706
SPAR	0.2073 ±0.0357	0.3068 ±0.0330	0.2232 ±0.0344	0.3309 ±0.0610	0.2554 ±0.0411	0.5254 ±0.0462